higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

MATHEMATICS
(First Paper)
NQF LEVEL 2

5 NOVEMBER 2019

This marking guideline consists of 10 pages.

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INSTRUCTIONS TO MARKERS

1. Examiners are reminded that the subject Mathematics lends itself to a variety of methods and, therefore, several different alternative answers. Mark ALL mathematically correct answers.

2. The answers provided in this memorandum are, therefore, not exhaustive. Marks are allocated in principle and the markers must adhere to the mark allocation.

3. Please mark the questions and follow up, for example, an error in the first step of differentiation and a mistake in converting from surd to exponential do not imply that the student cannot differentiate.

4. Marks are allocated per step, but if a student omitted a step and there is evidence or reason that he/she can derive at the next step without calculation, then the mark must still be given to the student.

5. Take note that marks have been allocated for simplification in order to differentiate between your 80 percenters and 90 percenters.

6. Steps must be marked and not only the final answer. The number of ticks must correspond with the total. Write your ticks and crosses neatly and legibly in RED pen only.

7. If the answer does not look exactly the same as in the marking guideline, it could be an alternative method or a method that is not known to the marker. Please take a moment before you mark and verify the students' responses.

8. Mark ALL the questions.

9. Read ALL the answers carefully.

10. Follow the answers according to the numbering system used in this question paper.

11. Do NOT adjust students' marks to give them a pass mark.

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✓ **1 mark** √ **½ mark**

QUESTION 1

1.1 1.1.1 C✓ or real and rational number.

1.1.2 A✓ or $\frac{2}{-3a^5b^2}$

1.1.3 B✓ or $\frac{2}{5\sqrt[3]{x^2}}$

1.1.4 D✓ or $1,\dot{3}\dot{1}$

1.1.5 A✓ or $x \in (-3;5]$

(5 × 1) (5)

1.2 1.2.1 321,515151....

Let $x = 321,\dot{5}\dot{1}$

$100x = 32151,\dot{5}\dot{1}$ √

$\therefore 100x - x = 31830$

$\therefore \quad 99x = 31830$ √

$\therefore \quad x = \frac{31830}{99}$ √ $= \frac{10610}{33}$ √

(2)

1.3 1.3.1

$= \frac{5\sqrt{2}\,x + \sqrt{32}.\sqrt{x^2}}{\sqrt{81}\,x}$

$= \frac{5\sqrt{2}x \ + 4\sqrt{2}x}{9x}$ ✓

$= \frac{9\sqrt{2}x}{9x}$ ✓

$= \sqrt{2}$ ✓

(3)

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1.3.2

$$\frac{\sqrt{36x^2}}{x-\sqrt{5x^2}}$$

$$¿\frac{6x}{x-\sqrt{5}\sqrt{x^2}}¿\frac{6x}{x-x\sqrt{5}}¿\frac{6x}{x(1-\sqrt{5})}¿\frac{6}{(1-\sqrt{5})}\times\frac{1+\sqrt{5}}{1+\sqrt{5}}¿\frac{6(1+\sqrt{5})}{1-\sqrt{5}+\sqrt{5}-5}$$

$$¿\frac{6+6\sqrt{5})}{-4}$$ (3)

1.4 1.4.1 $(2x^{-3}\times 4x^3)^2 \times x^0=(8x^0)^2\times 1$

$=(8)^2=64$ (2)

1.4.2 $\frac{2^x+2^{x+3}}{2^x}$

$=\frac{2^x+2^x\times 2^3}{2^x}$

$=\frac{2^x(1+8)}{2^x}=9$ (2)

1.4.3 $\frac{(a^4)^{n+1}\times(a^2b)^{-3n}}{(a\ b)^{-2n}\times b^{-n}}$

$=\frac{(a)^{4n+4}\times(a^{-6n}b^{-3n})}{(a^{-2n}b^{-2n})\times b^{-n}}$

$=\frac{a^{-2n+4}b^{-3n}}{a^{-2n}b^{-3n}}=\frac{a^{-2n}a^4b^{-3n}}{a^{-2n}b^{-3n}}$

$=a^4$

Alternative:

$$\frac{(a^4)^{n+1}\times(a^2b)^{-3n}}{(ab)^{-2n}\times b^{-n}}¿\frac{(a)^{4n+4}\times(ab)^{2n}\times b^n}{(a^2b)^{3n}}¿\frac{a^{4n+4}a^{2n}b^{2n}b^n}{a^{6n}b^{3n}}$$

$$¿a^{4n+4+2n-6n}b^{2n+n-3n}$$

$$¿a^4$$ (3)

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1.5 1.5.1

$y = \frac{3}{x} + 5$

$\therefore y - 5 = \frac{3}{x}$

$\therefore x(y-5) = 3$ ✓

$\therefore x = \frac{3}{y-5}$ ✓

Alternative

$y = \frac{3}{x} + 5$

$yx = 3 + 5x$

$yx - 5x = 3$

$\therefore x(y-5) = 3$ ✓

$\therefore x = \frac{3}{y-5}$ ✓

(2)

1.5.2 $At\ y = 10$

$x = \frac{3}{10-5} = \frac{3}{5}\ or\ 0,6$ ✓

(1)

1.6 1.6.1

$T_{15} = a + (n-1)d$

$T_{15} = 6 + (15-1)3$ ✓

$T_{15} = 48$ ✓

(2)

1.6.2

$S_n = \frac{n}{2}\left[2a + (n-1)d\right]$

$S_{100} = \frac{100}{2}\left[2(6) + (100-1)3\right]$ ✓

$= \frac{100}{2}\left[2(6) + (99)3\right]$

$= 50(309)$

$= 15450$ ✓

(2)

1.6.3

$T_n = a + (n-1)d$

$138 = 6 + (n-1)3$ ✓

$138 - 6 = 3n - 3$

$132 + 3 = 3n$ √

$3n = 135$

$\therefore n = 45$ ✓

(3)

[30]

QUESTION 2

2.1 2.1.1 B✓ or $y = -2$

2.1.2 D✓ or $y \in (-2\,;+\infty)$ where $y \in R$

2.1.3 D✓ or The name of $g(x)$ is an exponential graph.

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2.1.4 A✓ or increasing

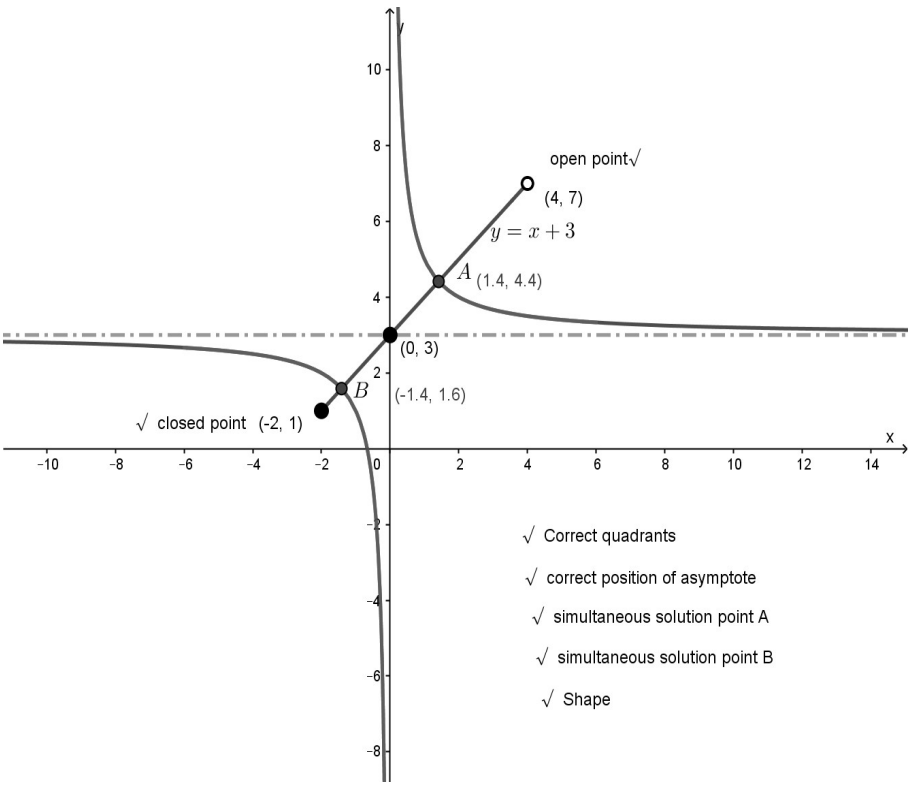
2.1.5 B✓ or Domain:$\{x|-2\leq x<4\,;\,x\in R\}$ and range:$\{y|-6\leq y<3\,;\,y\in R\}$

2.1.6 D✓ or $x=0$

2.1.7 C✓ or vertical transformation in the graph upwards.

(7 × 1) (7)

2.2 2.2.1 & 2.2.2



(7)

2.2.3 $y=x+3$✓
$y=-x+3$✓ (2)

2.2.4 If "a" is negative, the graphs will be increasing graphs ✓,
or "a" is negative the graphs will be quadrant 2 and 4 (1)

2.3 2.3.1 $q=3$✓
$h(x)=ax^2+3$
$1=a(1)^2+3-2=a$ ✓
$\therefore\ h(x)=-2x^2+3$ √ (3)

2.3.2 Parabola (1)

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2.3.3 Range $=\{y : y \in R ; -\infty < y \leq 3\}$ ✓✓
or Range $y \in (-\infty ; 3]$ where $y \in R$ ✓✓
or $y \leq 3$; $y \in R$ ✓✓ (2)

2.3.4 $h(x)$ is a function .✓
Reasons:
x values do not repeat ✓
For every x value, there is only one y value.
Vertical lines cut only once (2)
[25]

QUESTION 3

3.1 3.1.1 $(2x-3)^2$
$= 4x^2 - 6x - 6x + 9$ ✓
$= 4x^2 - 12x + 9$ ✓

3.1.2 $(2x-y)(4x^2+2xy+y^2)$
$= 8x^3 + 4x^2y - 4x^2y + 2xy^2 - 2xy^2 - y^3$ ✓
$= 8x^3 - y^3$ ✓

(2 × 2) (4)

3.2 3.2.1 $12a^2 - 3b^2$
$= 3(4a^2 - b^2)$ ✓
$= 3(2a-b)(2a+b)$ ✓ (2)

3.2.2 $14 - 5p - p^2$
$= -(p^2 + 5p - 14)$ ✓
$= -(p+7)(p-2)$ ✓
Alternative:
$14 - 5p - p^2$
$= (p+7)(2-p)$ or ✓✓$(7+p)(2-p)$ (2)

3.2.3 $(2x^3 - 4x^2) + (-2 + x)$
$2x^2(x-2) + (x-2)$ ✓
$(x-2)(2x^2+1)$ ✓
Alternative:
$(2x^3 + x) - (4x^2 + 2) = x(2x^2+1) - 2(2x^2+1) = (x-2)(2x^2+1)$
✓ ✓ (2)

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3.3 $\dfrac{4(2a+1)-5(4a-2)}{20a}$

$= \dfrac{8a+4-20a+10}{20a}$ √

$= \dfrac{-12a+14}{20a}$ √

$= \dfrac{2(-6a+7)}{20a}$ √

$\dfrac{(-6a+7)}{10a}$ √

(2)

3.4 3.4.1 $x^2+6x=16$

$x^2+6x-16=0$ √

$(x+8)(x-2)=0$ √

$x+8=0 \ or \ x-2=0$ √

$\therefore x=-8 \quad or \quad x=2$ √

(2)

3.4.2 $\dfrac{2}{x}=\dfrac{-4}{x+5}$

$2(x+5)=-4x$ √

$\therefore 2x+10+4x=0$ √

$\therefore 6x=-10$ √

$\therefore x=\dfrac{-10}{6}=\dfrac{-5}{3}$ √

(2)

3.4.3 $2^{x-4}=1$

$2^{x-4}=2^0$ √

$\therefore x-4=0$ √

$\therefore x=4$ ✓

Alternative:

$2^{x-4}=1$

$2^x 2^{-4}=1$ √

$\therefore 2^x=\dfrac{1}{2^{-4}}$ √

$\therefore 2^x=2^4$

$\therefore x=4$ ✓

(2)

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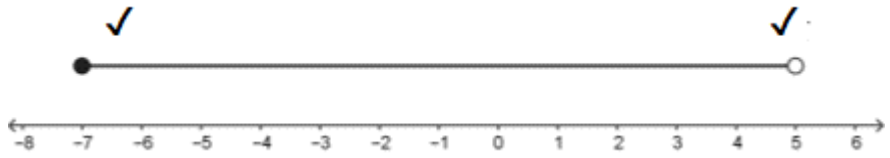
3.5 3.5.1 $-2 < -x + 3 \leq 10$

$-5 < -x \leq 7$ √

$\therefore 5 > x \geq -7$ √

$-7 \leq -x < 5$ ✓ (2)

3.5.2



(2)

3.6

$x + 2y = 3$.....eq1

$3x - 2y = 5$.....$eq2$

From eq1:

$x = 3 - 2y$ √substitute in eq2

$\therefore 3(3 - 2y) - 2y = 5$ √

$\therefore 9 - 6y - 2y = 5$

$\therefore 8y = 4$

$\therefore y = \frac{4}{8} = \frac{1}{2}$ ✓

$\therefore x = 3 - 2\left(\frac{1}{2}\right)$

$\therefore x = 2$ ✓

Alternative:

$x + 2y = 3$.....eq1

$3x - 2y = 5$.....$eq2eq1 + eq2$

$\therefore 4x = 8$ ✓

$\therefore x = 22y = 3 - x2y = 3 - 2 = 1$

$\therefore y = \frac{1}{2}$ ✓

Alternative:

$x + 2y = 3$.....eq1$\times -3$

$3x - 2y = 5$.....$eq2$

$-3x - 6y = -9$..√.....$eq3$

$eq2 + eq3$

$\therefore 0 - 8y = -4$ √

$\therefore 8y = 4$

$\therefore y = \frac{4}{8} = \frac{1}{2}$ ✓

$x = 3 - 2y$

$\therefore x = 3 - 2\left(\frac{1}{2}\right)$

$\therefore x = 2$ ✓

(3)

[25]

✓

QUESTION 4

4.1 4.1.1 G✓ or Wants

4.1.2 H✓ or Stokvels

4.1.3 F ✓ or Savings

4.1.4 E✓ or Hire Purchase Agreement

4.1.5 A✓ or Bank Fees

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(5 × 1) (5)

4.2 4.2.1 A $-R200$ is unfavourable, because she received less money than she budgeted for. ✓ (1)

4.2.2 $A = 7300$ ✓ $B = R850$ ✓ (2)

4.2.3 Actual income less actual expenses
$= R7\,325 - R5\;650 = R1\,675$

Yvette has a surplus of $R1\,675$ ✓ ✓. (2)

4.2.4 Gym membership and long-term investments are fixed expenses. (1)

4.3 4.3.1

$$A_t = A_o(1 + \frac{rt}{100})$$

$$A_t = 1700(1 + \frac{7(3)}{100}) \checkmark$$

$$A_t = R2057 \checkmark$$

(2)

4.3.2

$$A_t = A_o(1 + \frac{r}{100})^t$$

$$A_t = 1700(1 + \frac{7}{100})^3 \checkmark\checkmark$$

$$A_t = R2082,57 \checkmark$$

(3)

4.3.3 Compound interest gives the best growth. (1)

4.4

$$A_t = A_o(1 + \frac{r}{100})^t$$

$$A_t = 1730(1 + \frac{6,3}{100})^3 \checkmark\checkmark$$

$$A_t = R2078,00 \checkmark$$

(3)
[20]

TOTAL: 100

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